PUBLIC SAFETY AND HOMELAND SECURITY BUREAU

CYBERSECURITY BEST PRACTICES GUIDANCE FOR COMMUNICATIONS SERVICE PROVIDERS

October 2026

OVERVIEW

This guide details the current communications sector threat landscape, best practices for communications service providers, the stages of a successful incident response plan, and additional cybersecurity reference resources.

Additional best practices can be found in the Bureau's Oct. 23, 2026 Public Notice.

THREAT LANDSCAPE

Communications sector entities face growing danger from criminal groups, foreign state actors, hacktivists, and insider threats.

Threat Actors Target Critical Infrastructure

The communications sector is a key component of U.S. critical infrastructure and must be resilient to cyberattacks intended to disrupt communications services or exfiltrate sensitive commercial or personal data. Communications sector entities, including communications service providers, face growing dangers from criminal groups, foreign state actors, hacktivists, and insider threats.

Telecommunications infrastructure is of particular interest to threat actors, as it is both a target and an attack path: the Internet is the "highway" through which cyberattacks are conveyed, and the telecommunications sector undergirds other critical lifeline functions, such as energy, water, transportation, and finance.

Access Vectors and Ways to Cause Harm

There are multiple ways a threat actor can gain access to communications networks to cause operational disruptions, hijack systems or services, or exfiltrate data.

Threat actors might exploit unpatched software or equipment, engage in social engineering techniques like impersonating a help desk employee or contractor to gather credentials or attempt to gain access through a compromised link in an e-mail. Once access is acquired, malicious actors can use ransomware, a type of malicious software, to encrypt a target system's data before demanding a ransom or extorting the victim to decrypt it.

Hijacked Broadcasts

The effects of a successful cyberattack can have devastating consequences for a broadcast station.

Unauthorized access could lead to interfering with broadcasts, hijacking transmissions, or manipulating Emergency Alert System encoders or decoders.

Disrupted Communications and Data Loss

Risks to telecommunications service providers include data loss, interruption in service continuity, and disruption of emergency communications.

Vulnerabilities in small providers may create risks for the entire sector. This makes robust planning, coordination, and collaboration critically important.

BEST PRACTICES

Providers should implement a set of best practices to better mitigate cybersecurity risks.

Develop a Cybersecurity Risk Management Plan

A coherent and thoughtful cybersecurity risk management plan can help shift a provider's security posture from reactive to proactive, allowing it to prioritize threats and make smarter, more cost-effective security investments.

Ensure Stakeholder Buy-in and Establish Rules of Behavior.

Preparing for cyberattacks takes buy-in from all internal stakeholders.

This includes everyone from technical teams to senior leadership, and may also include external parties, such as trusted vendors, cyber insurance providers, outside counsel, and marketing/public relations firms.

To guard against ransom, extortion, and harassment, a cybersecurity risk management plan may include implementing certain practices to reduce the target attack surface for threat actors, including restricting help desk password resets without involving a person's manager (to prevent help desk misuse) or prohibiting employees from saving personal contacts on work infrastructure (to prevent the threat actor from using them as leverage).



Consult the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) to better understand, assess, prioritize, and communicate cybersecurity risk management strategy.

Assess and Understand Supply Chain Risk.

When developing a risk management plan, providers should assess their supply chain risks, including those posed by vendors and other third parties, as attackers are increasingly moving upstream from the communications service provider to target customers or other third parties whose systems are connected with those of the provider.

Establishing a line of communication with vendors can facilitate collaboration when situations arise that affect third party systems and require rapid restoration of the network to a trusted state.

Conduct a Risk Assessment and Prioritize Critical Assets

Prioritizing protections based on risk can guide your organization's approach in bite-sized, attainable steps that are not overwhelming.

Inventory Network Assets.

Create an inventory to understand what assets are in your network, who is responsible for them, and what their potential attack surfaces and vulnerabilities may be. A visual diagram or map of your systems can help you determine what is critical, what is exposed, and where there are dependencies.

Assess Risk.

Consider the likelihood of a compromise and the amount of harm that could arise from that compromise for each asset. If the amount of potential harm is very high, that asset is likely one that should be protected at the outset.

If an asset is highly vulnerable, but its compromise would not expose anything critical to your operations, address other assets first.

More Ways to Keep Up with Potential Vulnerabilities and Mitigate Risk:

- **Check CISA's Known Exploited Vulnerabilities Catalog** to see if there are already-identified exploitable vulnerabilities that your organization can address.
- **Use an automated vulnerability scanning system**. Routinely assess and reassess your controls, and build in effective protections over time.

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Practice Basic Cyber Hygiene

All organizations can implement these simple and accessible cybersecurity protections quickly with high impact. These practices focus on preventing attackers from breaching the outer boundary of a network (known as traditional "perimeter-based network security").

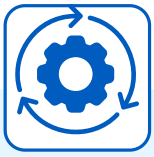


Identity Management, Authentication, and Access Controls

Begin with the "low hanging fruit," such as password security.

Authenticate. Ensure that users, services, and hardware are authenticated securely by changing default and test/configuration passwords, enabling multi-factor authentication (MFA), and ensuring passwords and passphrases are complex.

Access Controls. When applying basic access controls, consider what information employees should or do have access to and implement firewalls to protect against malicious traffic and restrict sensitive information to only those employees who need it.



Software and Hardware Updates

Software and hardware should be kept up to date or replaced.

Critical Software/Equipment. Prioritize installing critical software and equipment – especially for systems that are exposed to the public Internet.

Test Updates. Software updates should be tested before deployment to ensure they are safe and that the software operates as expected prior to use, and patches should be implemented as needed.

Replace. Consider replacing older/legacy devices that can no longer be updated or patched the device. The cost of a compromise could be greater than the cost of replacement.



Employee Training

Train employees and contractors on proper cyber hygiene practices.

Basic Hygiene. Employees should know how to perform the basic cyber hygiene tasks (e.g., avoiding phishing scams and backing up data consistent with company policies).

Recognition and Fast Outreach. Ensure employees know how to recognize common attacks and who to contact should problems arise. They should also be encouraged to reach out to those contacts as soon as possible to enable the organization to get ahead of a cyber problem or avoid it.



Data Backups

For data security, use encryption to protect the confidentiality and integrity of network communications.

Regularly Backup and Test. Regularly back up your organization's data and test your backups to ensure they are reliable.

Store Backups Offline. Consider securely storing backups offline or offsite for additional protection.

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Scale Up: Implement a Zero-Trust Strategy

A zero trust strategy accounts for potential "insider threats" by assuming the adversary is already inside the network. Restrict accounts, programs, and systems to the bare minimum permissions (least privilege) needed for a user to do their job effectively.

Network segmentation, or a physical or architectural approach that divides a network into multiple segments, is a pillar of zero trust that works alongside least privilege. Segmentation can make it harder for threat actors to access key data. For example, a firewall can help separate essential systems and prevent a cyber threat actor from moving laterally from one part of a network to another, causing widespread damage. When segmenting your network, think about the service provided by each specific function (e.g., corporate/backend versus customer-facing) and whether they need to be connected, and separate those functions accordingly, where possible.



For broadcasters, segmentation can help protect the EAS encoder/decoder from cyber threats. Because Internet connectivity is required to reach the Common Alerting Protocol (CAP) and for EAS servers to transmit EAS alerts, segmentation for EAS equipment can include creating a firewall that only allows the encoder/decoder to talk to CAP servers, denying all other inbound traffic.

A broadcaster should secure the communications pathway between the studio and the transmitter.

Continuously Monitor for Indicators of Compromise

Monitoring is key to knowing when something abnormal is happening in your network. Every time a user logs in, accesses a file, or makes changes to your system, they leave a digital record. Logging these records, collecting the logs in a central location, and reviewing the logs in real time, or near real time, can create a clear picture of normal, baseline behavior. Having this baseline allows for deviations or unauthorized behavior to be quickly detected.

Your organization can enhance its system monitoring by reducing the number of connections between critical systems, with the added benefit of helping your organization save on monitoring expenses.



INCIDENT RESPONSE

Develop a Cyber Incident Response Plan

Make a cyber incident response plan part of your overall cybersecurity risk management strategy, and practice implementing the plan in advance. Your response plan should cover the lifecycle of a cyber incident: All business functions (e.g., security, IT, legal, communications) should be included in your response plan.

DETECTION > CONTAINMENT > INVESTIGATION > MITIGATION > REMEDIATION > RESTORATION

Establishing a core cyber response team can help ensure that each function's role and responsibilities are clear to enable a rapid response. What your organization uncovers during a cyber incident will change over time, so develop a plan that anticipates activating additional teams as needed.



The plan may only need to be a few pages long with a quick start guide and a flow chart of what do to after detecting a potential cyber incident, **but it should include triggers to escalate decision-making as soon as there is suspicion of a cyber incident**. Waiting for confirmation of an intrusion may be too late to effectively contain and mitigate a threat actor.

Contain, Investigate, Mitigate

After a cyber incident has been detected, responsive action should prioritize containment (e.g., resetting credentials, isolating affected systems, implementing machine-level containment, blocking malicious traffic).

An **investigation** into the root cause of the incident is also important to find the attacker's entry point, track lateral movement, and identify any data that may have been exfiltrated or compromised. Taking an entire machine out of service may address operational concerns, but failing to uncover the root cause could lead to a "whack-a-mole" issue of continuous breaches.



During **incident response**, providers should keep their risk thresholds in mind to ensure that containment and mitigation actions are taken in order of urgency. Determining and documenting what went wrong in a post-incident forensic report can help businesses and public safety authorities prepare for similar attacks in the future.

Prior collaboration and coordination can improve the effectiveness of the response, which can in turn increase the speed of recovery.

INCIDENT RESPONSE

Remediate and Restore

Recovering from a cyber incident includes restoring assets affected by the incident and returning to normal business operations quickly to reduce the impacts of the incident on your organization's systems and the public.

Communications—internally, externally, and with the government—during incident response are vitally important. Information that is shared should be as complete and accurate as possible.

- The initial set of facts that the response team learns about the attack is unlikely to be the same as the final set of facts, so your organization's leadership may have to make decisions with imperfect information.
- Recovery should also inform hardening activities or modernization plans to address any residual risk to protect against the incident happening again.

Federal Assistance

Communications service providers should reach out to federal authorities to report successful cyberattacks and seek assistance as soon as they become aware of an incident.



The FBI or CISA can assist in tactical monitoring by sharing relevant indicators of compromise, if available.

- Companies can report an incident to the FBI through their local FBI field office, or by submitting a report through the **FBI's Internet Crime Complaint Center**.
- Communications service providers can also share information with **CISA through its secure cyber incident reporting portal**.

The FCC Operations Center (202-418-1122 or FCCOPCenter@fcc.gov) is available 24/7 and can also assist with coordination.

Ransomware Attacks

Ransomware attacks a common threat facing providers, especially broadcasters and small-/medium-sized telecommunications service providers.

In many ransomware attacks, the threat actor holds the victim's systems for ransom, threatening their ability to provide communications services. **One of the most important things a provider can do is prospectively ensure it has sufficient backups to recover and restore operations.**

After some initial investigation, teams may need to consider whether or how to engage with a ransomware threat actor. **Having a practiced incident response plan can help teams avoid shutting down a system prematurely, which could have unintended operational consequences and alert a threat actor that it has been detected.**

Providers should prepare to engage with their local FBI field office and other third parties such as their cyber insurer or auditor when considering whether to engage in negotiations with a ransomware threat actor.



ADDITIONAL INFORMATION

Cybersecurity Resources

Join communities and build relationships that enable your organization to learn about active cybersecurity threats and best practices to secure your systems against those threats.

The Communications Information Sharing and Analysis Center (COMM-ISAC) is a forum for organizations across the telecommunications industry and government to exchange threat information and share success stories of effective security implementations.

Various communications industry associations and related groups can also offer resources and best practices. No matter the size or sophistication of your organization, there are resources out there to help you prepare in case an incident happens, for the sake of those who rely on your operations. **See small business resources** from the FCC, NIST, CISA, and Cybershare.

The FBI encourages organizations to get in touch with your local field office as soon as possible if a breach occurs so they can help and share information that can assist in your organization's situation.

This guidance draws on panel discussions and expert presentations at the Commission's May 14 & 15, 2026 cybersecurity workshops.

Broadcaster Workshop	Public Notice	Recording
Telecommunications Carrier Workshop	Public Notice	Recording

More Resources from the FCC

CSRIC IV's Report on Cybersecurity Risk Management and Best Practices (2015)

Cybersecurity Planning Guide

Cybersecurity for Small Businesses

More Resources from CISA

Cybersecurity Risk Assessment Guide

Known Exploited Vulnerabilities Catalog

Cybersecurity Best Practices